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Efficient Computer Vision
Models for Silkworm Feeding Prediction
and Habitat Analysis

Università degli Studi di Roma **La Sapienza**
Computer Vision 2024-2025

Outline

Problem statement

Context: Silkworm rearing is traditionally labor-intensive and prone to human error

Objective: Automate the monitoring process of silkworms' rearing

Tasks:

- Supervised Binary Classification
- Unsupervised Semantic Segmentation

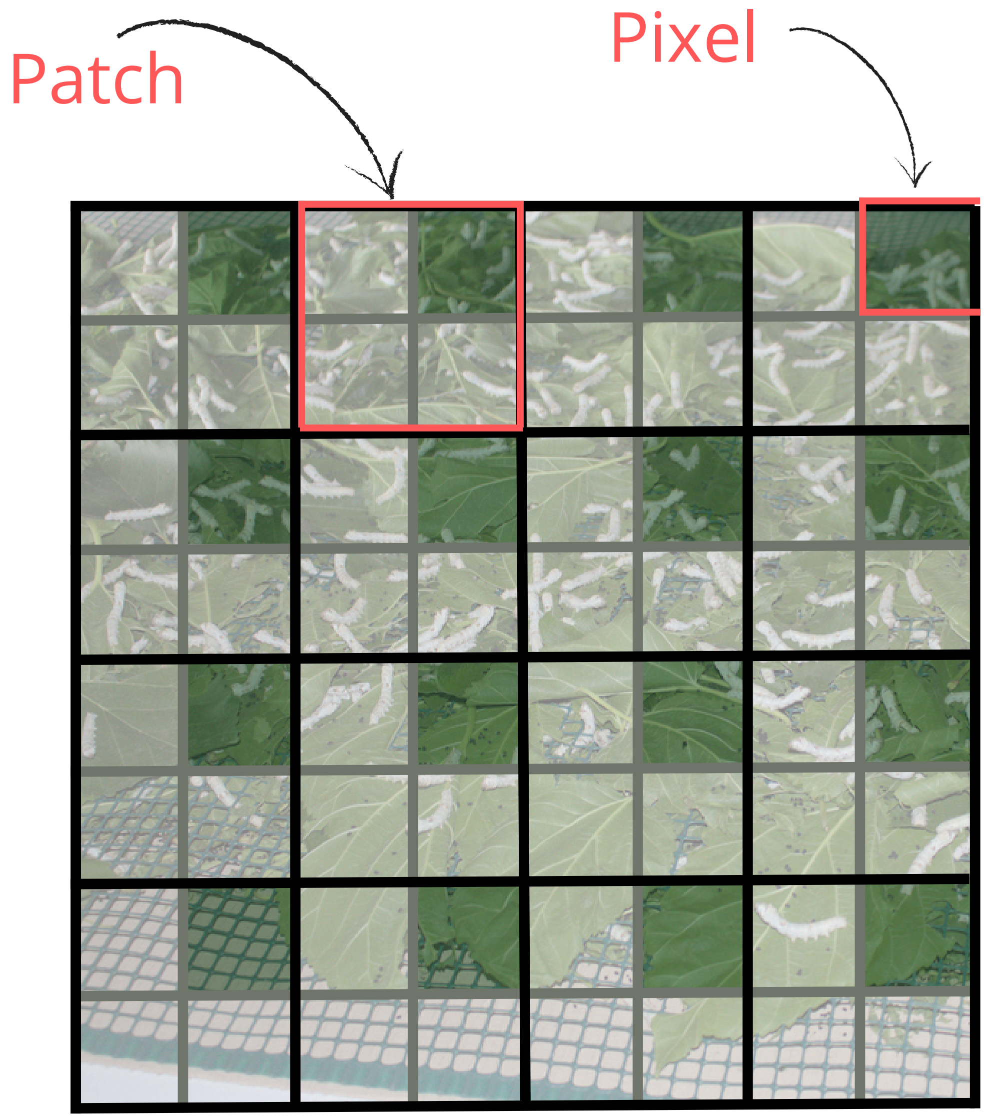
Feeding or not feeding ?

where is the worms, the leaves and the background?

Deep Learning for Vision

	CNN	ViT
Efficiency	extract local features	extract global features
Necessity	Deepy	many parameters

Hybrid Architectures merge the advantages of CNNs and Transformers



- **2DConv 3x3:** every pixel sees his 8 nearest neighbours
- **Self Attention:** every patch sees the others
- **Result:** each pixel sees both the local and the global context

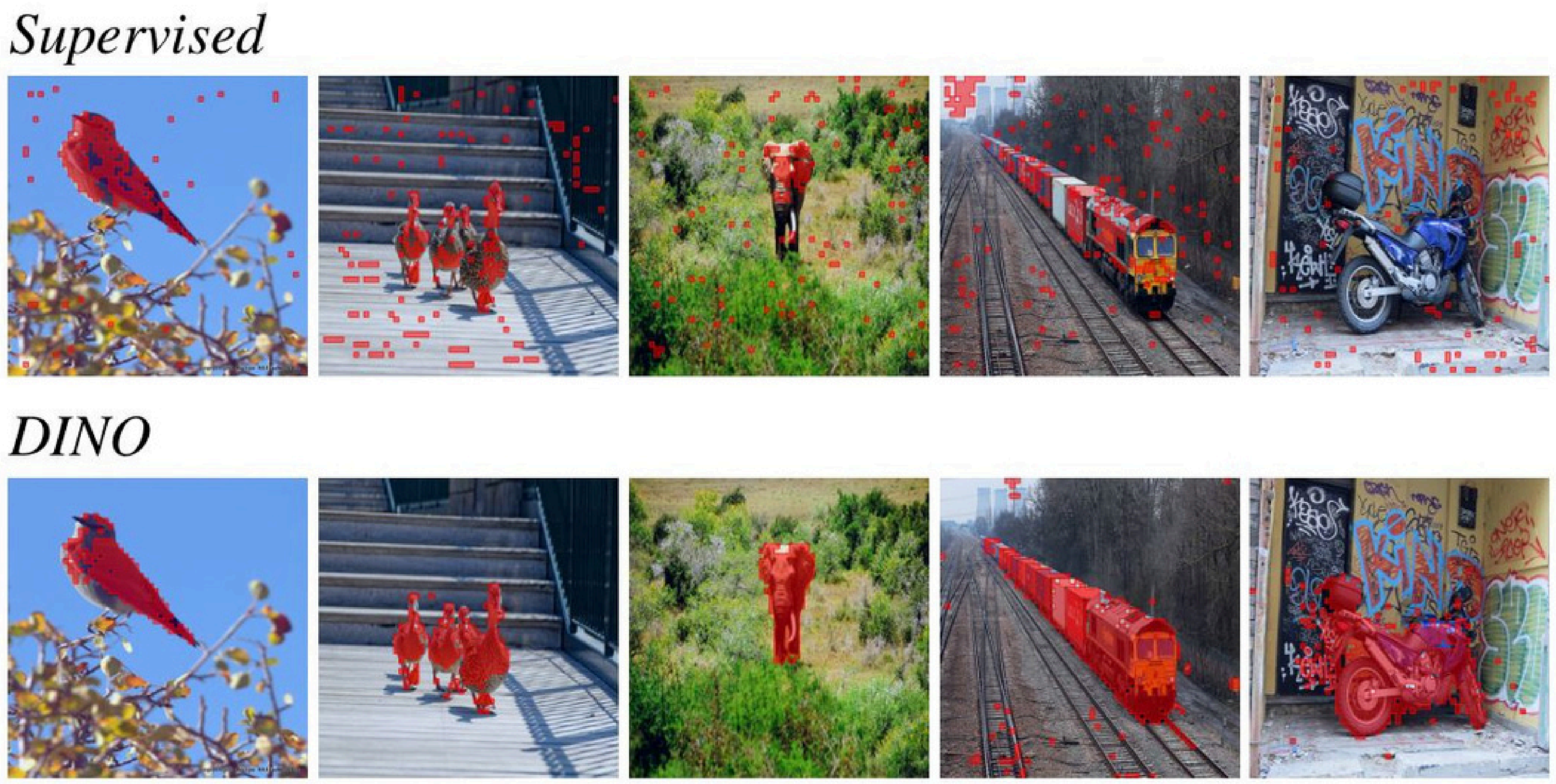
Unsupervised Semantic Segmentation

CNNs/Transformers used as feature **extractors**, followed by **clustering** on similar features.

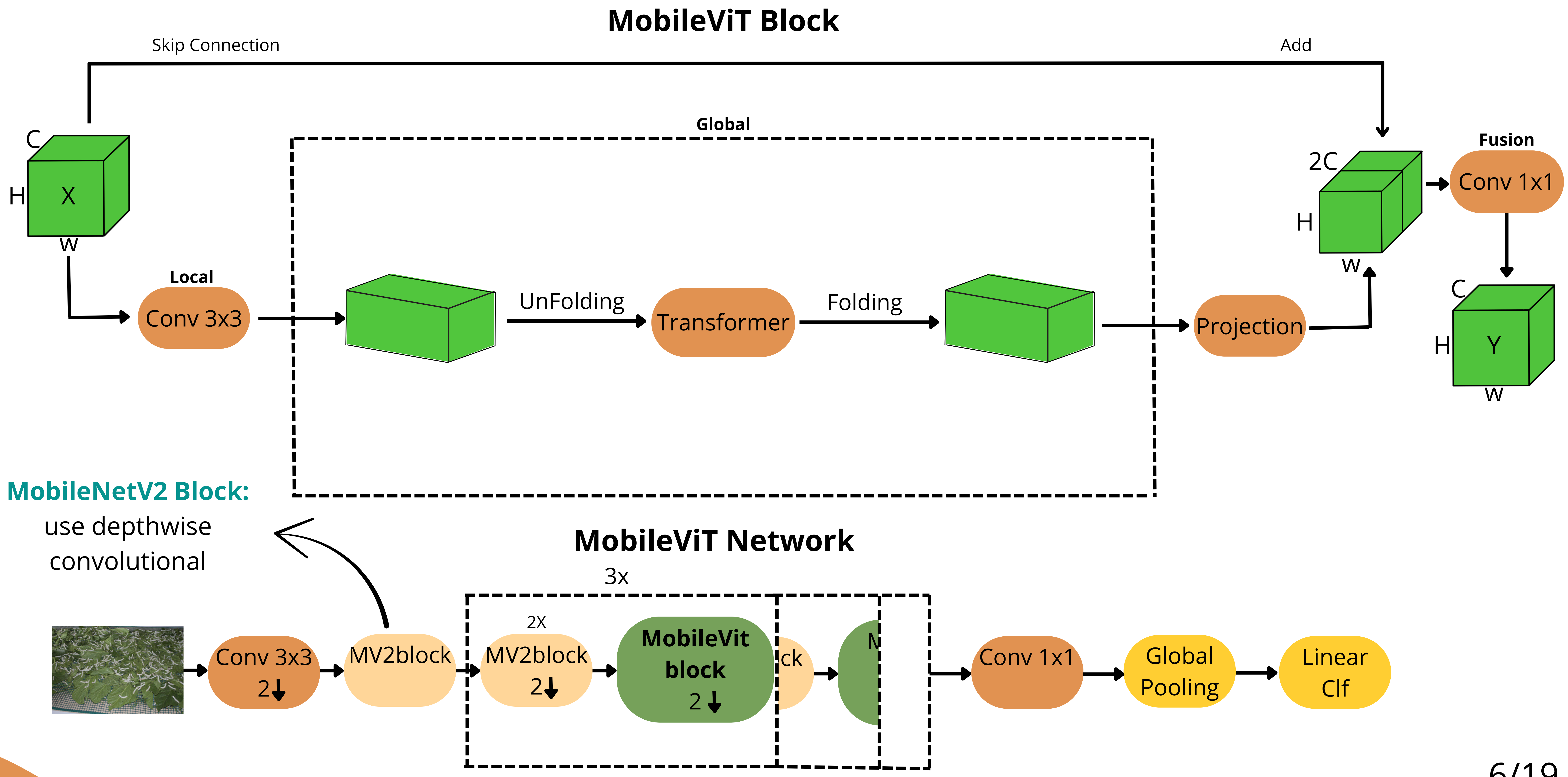
It's not clear how **supervised** models learn features

Self Supervised models highlight **semantic structures** such as edges and objects.

DINO (**Di**stillation with **no** labels) is trained with self distillation approch



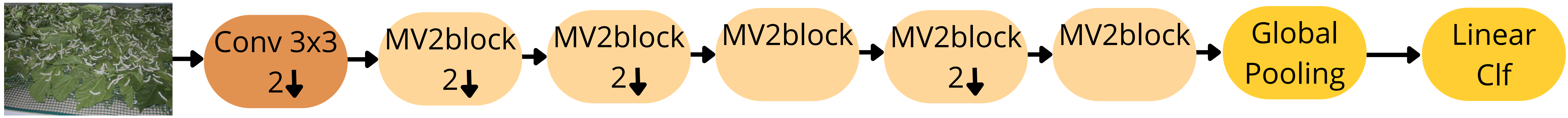
Proposed Method



Another Network

Let’s also consider a network that doesn’t use the Transformer Architecture

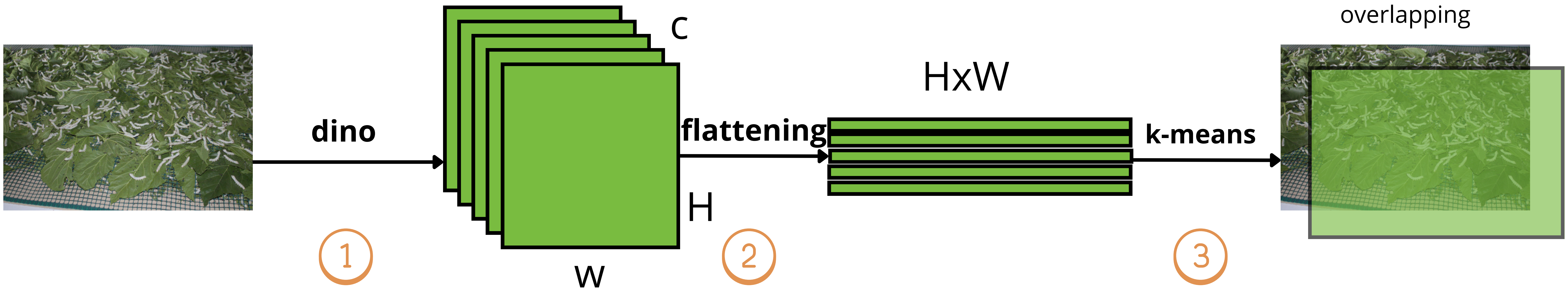
MiniCNN



Number of parameters for various configuration of models used in the project

Model	N_parameters
MiniCNN	948.130
MobileViT-xxs	1.040.898
MobileViT-xs	2.068.858
MobileViT-s	5.327.074

DINO as feature extractor + k-means



1 **Extracting** the feature from an intermediate layer of pretrained DINO

2 **Flattening** the spatial feature, prepare of k-means

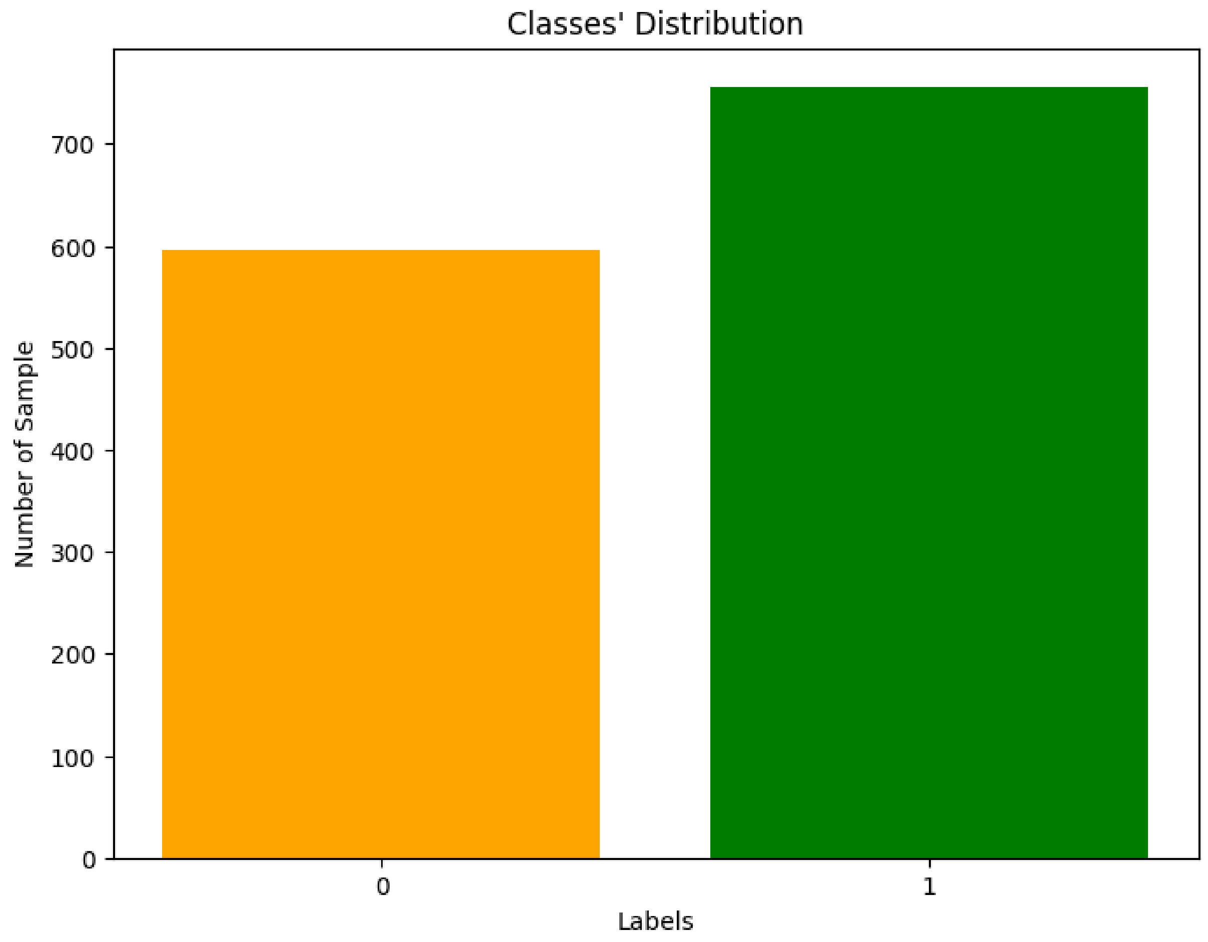
3
- **Clustering** the similar feature with k-means n_classses=3
- **Upsampling** the clusters on the image
- **Overlapping** with original image

Dataset

Let's have a look at the dataset to understand which strategies to execute

1351 Labeled Image

- Size: (1559,1038)
- 2 classes: feed/no-feed
- no-feeding:600 / feeding:751



0: no-feeding

1: feeding

Metrics

Classification

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{Precision} = \frac{TP}{TP + FP} \qquad \text{Recall} = \frac{TP}{TP + FN}$$

Avarage infer time per image [ms]

Semantic Segmentation

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i),\ b(i)\}}$$

$$\text{silhouette-score} = \frac{1}{N}\sum_{i=1}^{N} s(i)$$

Sperimental setup

Classification problem

Augmentation strategies:

Standard:
- RandomHorizontalFlip (50%)
- RandomRotation (10°)
- ColorJitter
- Normalize

Crop:
- RandomResizedCrop (60%-100%)
- + standard

Global Pooling strategies:

Attn pooling: Weighted sum of the spatial features
Avg pooling: AdaptiveAvaragePooling

Lightweight_models

Hyperparameters	N_epochs	Batch_size	Learning rate
Value	10	32	0.0001

Semantic segmentation Problem: Dino+ k_means

BackBone
Res-Net50

Average Performance of Lightweight Models Across Augmentation and Pooling Strategies

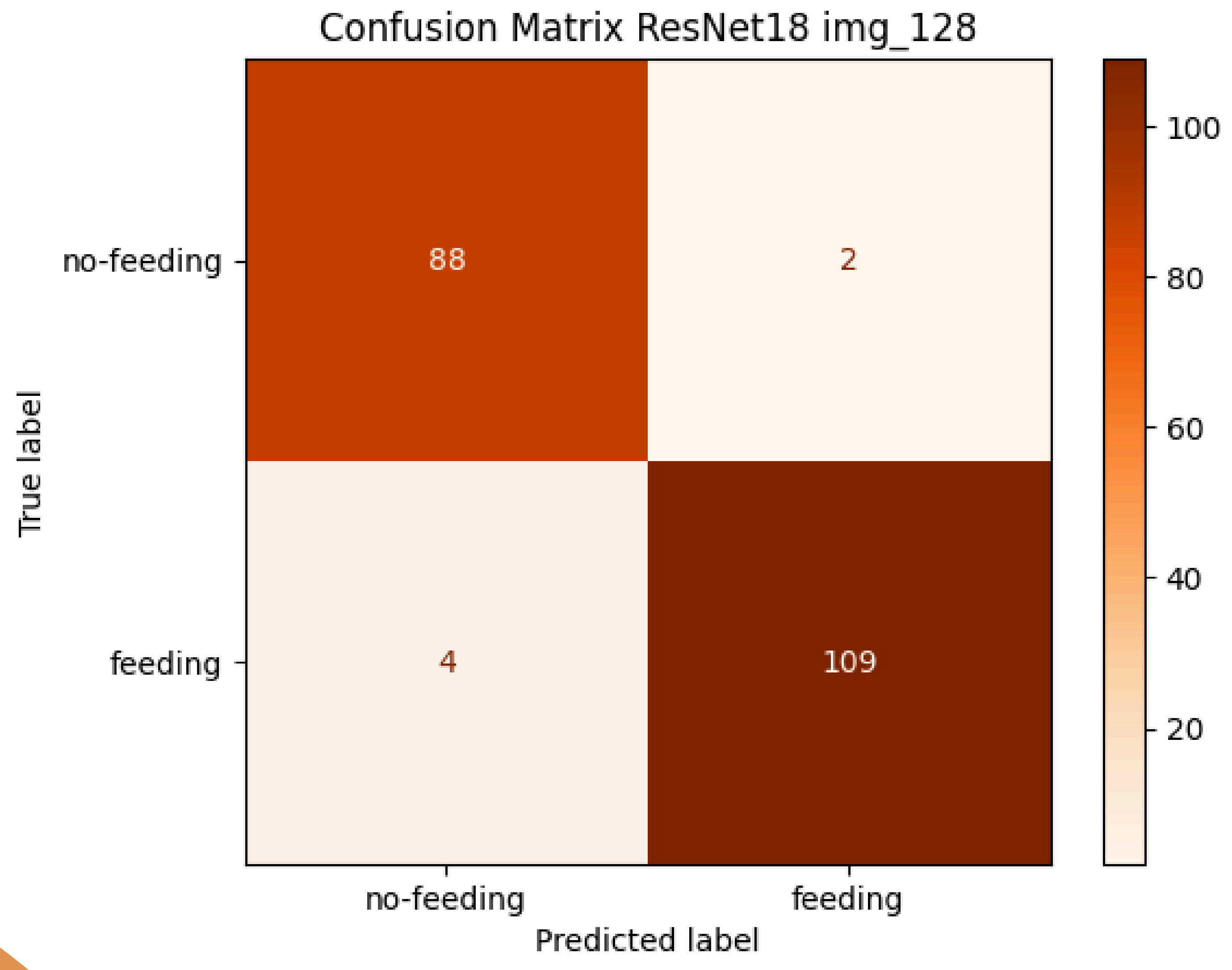
Augmentation	Pooling	F1_mean	F1_std	InferTime [ms]
standard_512	AdaAvgPooling	0,9534	0,0021	117,34
crop_512	Attn Pooling	0,9506	0,0104	119,97
crop_512	AdaAvgPooling	0,9489	0,0069	118,62
standard_512	Attn Pooling	0,9433	0,0109	118,97
standard_256	AdaAvgPooling	0,9431	0,012	26,05
crop_256	Attn Pooling	0,942	0,0134	26,61
standard_256	Attn Pooling	0,9383	0,0172	26,29
crop_128	Attn Pooling	0,9376	0,0129	4,84
standard_128	AdaAvgPooling	0,9368	0,007	5,365
crop_256	AdaAvgPooling	0,9335	0,0071	26,26
standard_128	Attn Pooling	0,9322	0,0112	4,745
crop_128	AdaAvgPooling	0,9258	0,0135	4,75

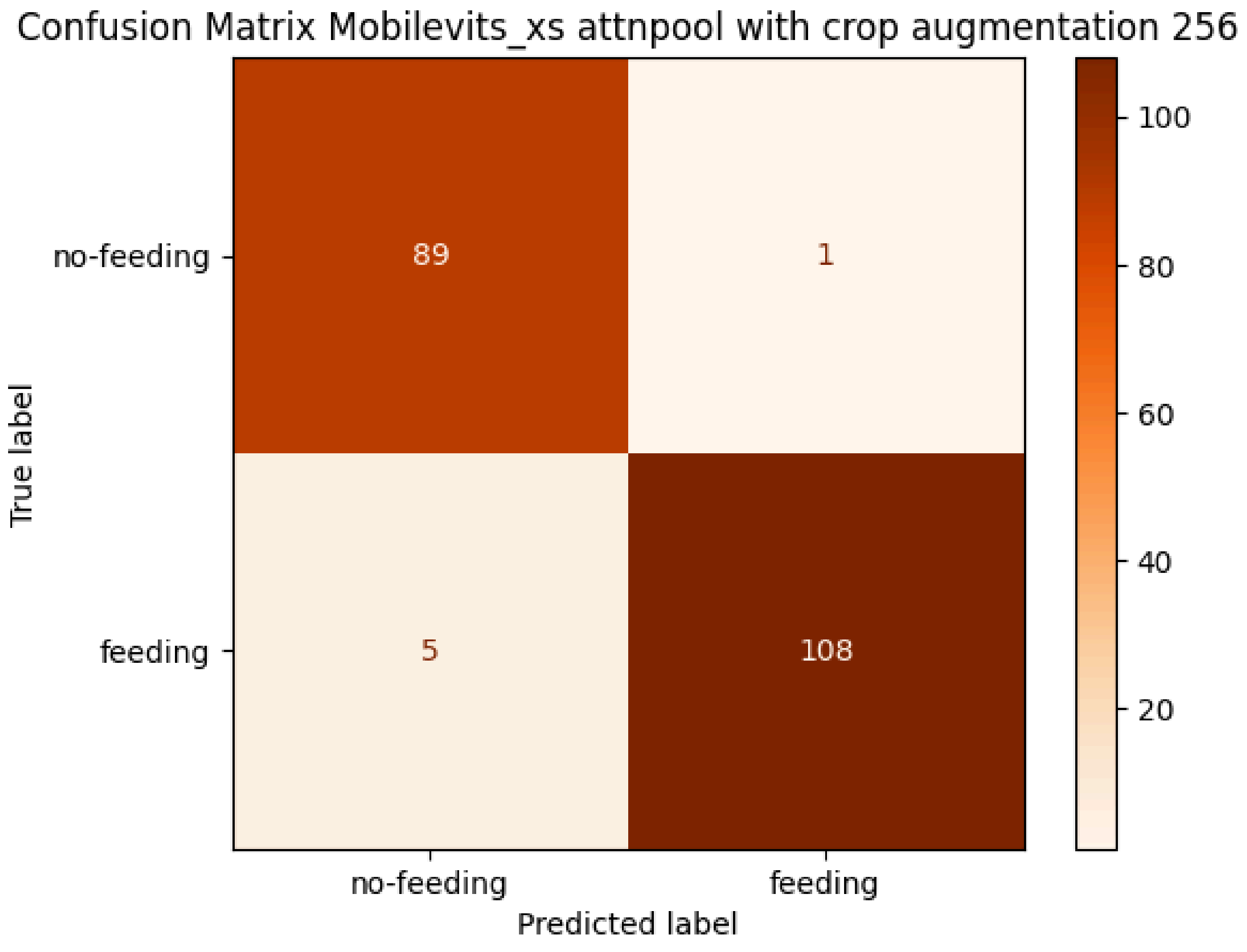
📌 Observations:

- Larger input sizes consistently lead to higher performances
- Attn pooling shows higher variance compared to average pooling → less stable

Lightweight Models vs Heavy Models

Model	Pooling	Augmentation	N_Parameters	Accuracy	F1-score	avg_infer_time[ms]
ResNet_18	AdaAvgPooling	standard_128	11.169.858	**0,9704**	0,9732	**84,72**
MobileViT-xxs	Attn Pooling	crop_256	1.040.898	0,9655	0,9692	11,03
MobileViT-xs	Attn Pooling	crop_256	2.068.858	**0,9704**	0,973	**29,19**
MobileViT-S	Attn Pooling	crop_256	5.327.074	0,9652	9.683	32,87





Future works

We want to minimize the false negatives for **welfare of silkworms**

Predict label → no-feeding

True label → feeding

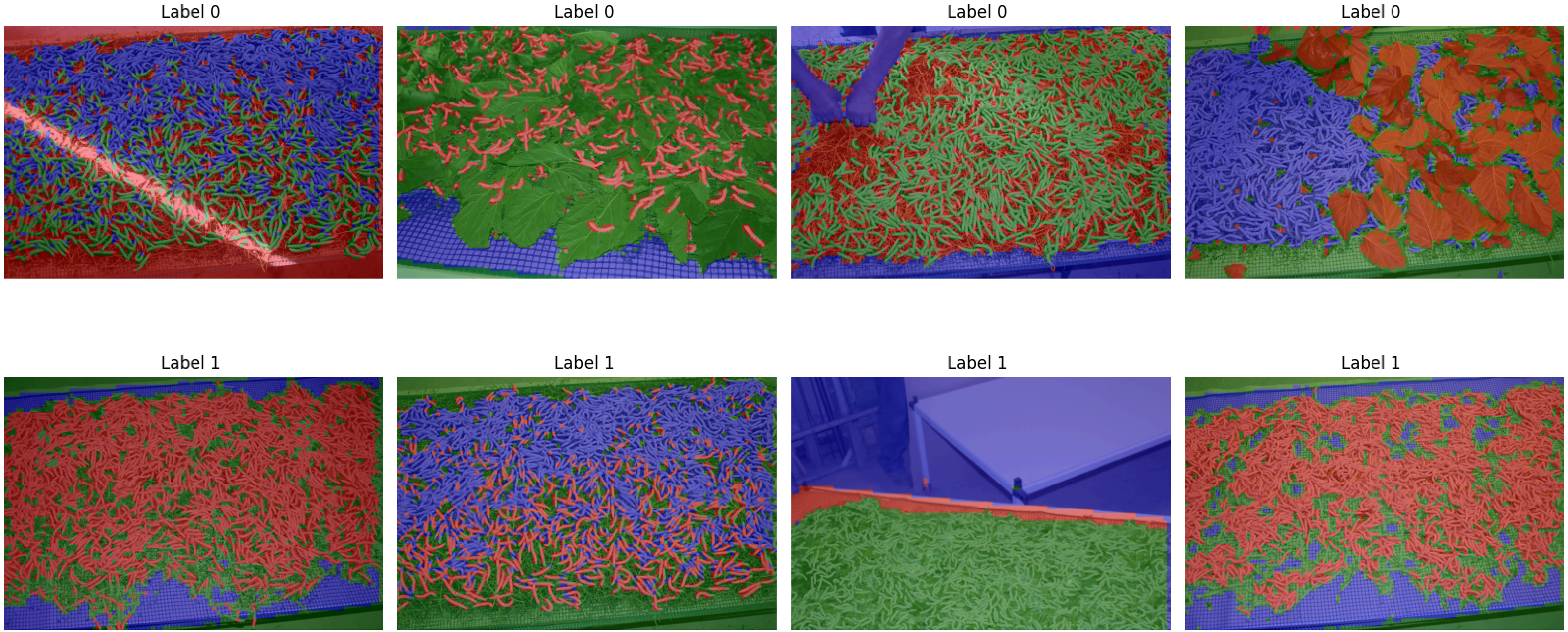
How to do this ?

Hyperparameter tuning to maximize the **recall**

$$\text{Recall} = \frac{TP}{TP + FN}$$

Results of semantic segmentation

Model output of some **representative examples:**



- Noisy backgrounds introduce segmentation confusion
- When a semantic class is missing from the image, the model overestimate the number of clusters

Future works

📌 Automitize the choice of n_clusters that **maximize** the **silhoutte score**

missing the leaves case

n_clusters=2 ?

DINO(ResNet50)_layer3 2clusters Sil_Score: 0.1667

Sil_score=0.17

n_clusters=3 ?

DINO(ResNet50)_layer3 3clusters Sil_Score: 0.0838

Sil_score=0.08

Let's test the method to ensure that it doesn't underestimate clusters when classes are well separated

All the classes are present case

n_clusters=2 ?

DINO(ResNet50)_layer3 2clusters Sil_Score: 0.0425

Sil_score=0.04

n_clusters=3 ?

DINO(ResNet50)_layer3 3clusters Sil_Score: 0.0753

Sil_score=0.07

Bibliography

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- MobileViT

Mehta, S., & Rastegari, M. (2021). MobileViT: Light-weight, General-purpose, and Mobile-friendly Vision Transformer
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- DINO

Caron, M., Misra, I., Mairal, J., Goyal, P., Bojanowski, P., & Joulin, A. (2021). Emerging Properties in Self-Supervised Vision Transformers. ICCV 2021.
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Thanks for your attention

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Appendix A

Best performances for input size: 128

Model	Augmentation	Pooling	F1_score	infer_time[ms]
MobileViT-S	standard_128	AdaAvgPooling	0,9397	8,58
MobileViT-S	crop_128	AdaAvgPooling	0,9238	7,3
MobileViT-S	standard_128	Attn Pooling	0,9386	7,43
MobileViT-S	crop_128	Attn Pooling	0,9447	7,54
MobileViT-XS	standard_128	AdaAvgPooling	0,9292	6,65
MobileViT-XS	crop_128	AdaAvgPooling	0,914	5,17
MobileViT-XS	standard_128	Attn Pooling	0,9156	5,11
MobileViT-XS	crop_128	Attn Pooling	0,9204	5,43
MobileViT-XXS	standard_128	AdaAvgPooling	0,9333	1,92
MobileViT-XXS	crop_128	AdaAvgPooling	0,9205	1,8
MobileViT-XXS	standard_128	Attn Pooling	0,9362	1,7
MobileViT-XXS	crop_128	Attn Pooling	0,9362	1,69
MINI_CNN	standard_128	AdaAvgPooling	0,9451	4,31
MINI_CNN	crop_128	AdaAvgPooling	0,9451	4,76
MINI_CNN	standard_128	Attn Pooling	0,9386	4,74
MINI_CNN	**crop_128**	**Attn Pooling**	**0,9492**	**4,7**

Appendix A

Best performances for input size: 256

Model	Augmentation	Pooling	F1_score	infer_time[ms]
MobileViT-S	standard_256	AdaAvgPooling	0,9573	36,51
MobileViT-S	crop_256	AdaAvgPooling	0,9304	37,08
MobileViT-S	standard_256	Attn Pooling	0,9286	37,79
MobileViT-S	crop_256	Attn Pooling	0,9381	38,25
MobileViT-XS	standard_256	AdaAvgPooling	0,9386	28,61
MobileViT-XS	crop_256	AdaAvgPooling	0,9298	29,47
MobileViT-XS	standard_256	Attn Pooling	0,9565	28,63
MobileViT-XS	crop_256	Attn Pooling	0,9369	30,13
MobileViT-XXS	standard_256	AdaAvgPooling	0,9292	11,23
MobileViT-XXS	crop_256	AdaAvgPooling	0,9442	11,02
MobileViT-XXS	standard_256	Attn Pooling	0,9196	11,49
MobileViT-XXS	crop_256	Attn Pooling	0,9316	11,15
MINI_CNN	standard_256	AdaAvgPooling	0,9474	27,88
MINI_CNN	crop_256	AdaAvgPooling	0,9298	27,5
MINI_CNN	standard_256	Attn Pooling	0,9487	27,26
MINI_CNN	**crop_256**	**Attn Pooling**	**0,9617**	**26,94**

Appendix A

Best performances for input size: 512

Model	Augmentation	Pooling	F1_score	infer_time[ms]
MobileViT-S	standard_512	AdaAvgPooling	0,9565	164,01
MobileViT-S	crop_512	AdaAvgPooling	0,9397	166,38
MobileViT-S	standard_512	Attn Pooling	0,9524	165,28
MobileViT-S	**crop_512**	**Attn Pooling**	**0,9652**	**172,57**
MobileViT-XS	standard_512	AdaAvgPooling	0,9524	131,8
MobileViT-XS	crop_512	AdaAvgPooling	0,952	134,01
MobileViT-XS	standard_512	Attn Pooling	0,9279	136,07
MobileViT-XS	crop_512	Attn Pooling	0,9407	137,4
MobileViT-XXS	standard_512	AdaAvgPooling	0,9524	60,28
MobileViT-XXS	crop_512	AdaAvgPooling	0,9558	58,66
MobileViT-XXS	standard_512	Attn Pooling	0,9437	59,47
MobileViT-XXS	crop_512	Attn Pooling	0,9487	56,93
MINI_CNN	standard_512	AdaAvgPooling	0,9524	113,27
MINI_CNN	crop_512	AdaAvgPooling	0,9483	115,44
MINI_CNN	standard_512	Attn Pooling	0,9492	115,06
MINI_CNN	crop_512	Attn Pooling	0,9478	112,98

Appendix B: Attention pooling

Let $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$ be the spatial feature map

1 Flattening and reshape $\mathbf{X}$ into $\mathbf{X}' \in \mathbb{R}^{B\times F\times C}, \quad \text{where } F = H \cdot W$

Each vector $\mathbf{x_{b,i}} \in \mathbb{R}^C$ represents the features at the i-th spatial position

Defining a shared MLP $f_\theta : \mathbb{R}^C \to \mathbb{R}$

2 $\forall i = 1, \ldots, F$ Compute the attention weight: $w_{b,i} = \text{softmax}(f_\theta(x_{b,i}))$

3 Compute the global feature: $z_b = \sum_{i=1}^{F} w_{b,i} \cdot x_{b,i}$ where $z_b \in \mathbb{R}^C$